

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456522.5235 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.047
Transit depth (Rp/Rs)^2	2.33 +/- 1.43 [%]
Orbital Inclination (inc)	88.24 +/- 2.9
Airmass coefficient 1 (a1)	1.877 +/- 0.0011
Airmass coefficient 2 (a2)	-0.46 +/- 0.34
Scatter in the residuals of the lightcurve fit is	0.031 %
Best Comparison Star	#2 - [15, 187]
Optimal Aperture	23.72
Optimal Annulus	173.97
Transit Duration (day)	0.0344 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.153 ± 0.047 vs 0.1326 (deviation 0.43σ)
Duration fit vs published	0.0344 d vs 0.0737 d (deviation 7.72σ)
Mid-time O-C	-3106.3 min
Depth SNR	3.3
Residual scatter	0.031 %

Light curve

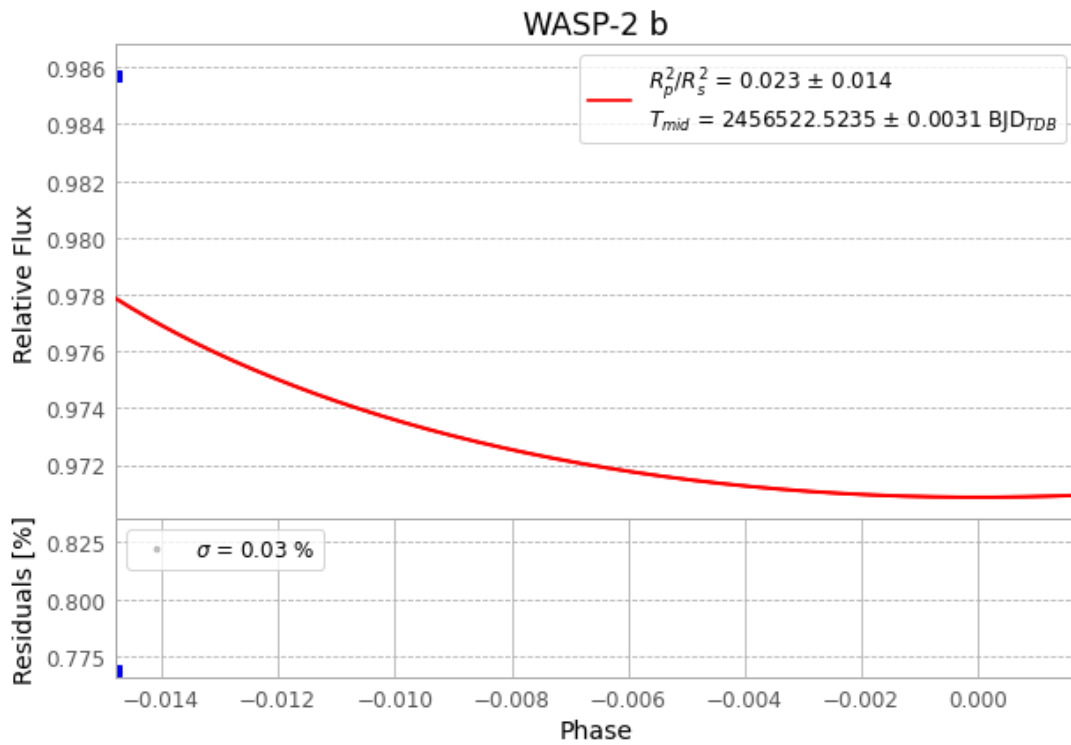


Figure 1 — FinalLightCurve_WASP-2 b_2013-08-19.png (SHA-256 620f3460e185b566...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [136, 333], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 775b86c3f83547e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-2130820031212.FITS → WASP-2130820062112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-130820.log (242050d8bd8e...), FinalLightCurve_WASP-2 b_2013-08-19.png (620f3460e185...), FinalLightCurve_WASP-2 b_2013-08-19.csv (73c2250efa18...)

Compute resources

Wall time 651.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 15:01Z.